Course: ENCN423/ENCN627
Module 10: Fuel tutorial

Question 1

A green hydrogen production system based on water electrolysis has the following characteristics:

Table 1: Electrolyser techno-economic parameters

Electrolyser installed capacity	50 MW_{el}
Electrolyser CAPEX	900 \$/kW
Hydrogen storage CAPEX	8 million \$
Fixed O&M cost	3% of CAPEX
Electrolyser lifetime	20 years
Discount rate	7%
Electrolyser efficiency	70% (LHV basis)
Annual full-load operating hours	4000 h/year
Electricity price	40 \$/MWh

Assuming a LHV value of 33.3 kWh/kg, answer the following questions:

(a) Calculate the levelised cost of hydrogen in \$/MWh and \$/kg?

(b) Determine the contribution (%) of the different costs components in the final levelised cost of hydrogen.

Question 2

Based on the previously calculated levelised cost of hydrogen, determine the levelised cost of methanol production ($/MWh) from an e-methanol synthesis plant and analyse the contribution of the different cost components to the final methanol production cost. The following input parameters are provided:

Table 2: Methanol plant parameters

Plant capacity	100 MW$_{fuel}$
Plant CAPEX	900 $/kW
Fixed O&M cost	4% of CAPEX
Plant lifetime	30 years
Discount rate	7%
Annual full-load operating hours	8000 h/year
Electricity consumption	0.03 MWh/MWh$_{fuel}$
CO_2 consumption	0.25 t/MWh$_{fuel}$
Hydrogen consumption	1.13 MWh/MWh$_{fuel}$
Electricity price	40 $/MWh
CO_2 price	120 $/$tCO_2$